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Steering Turret CORAL

- Phase B –

SW Interface Control Document

(Deliverable DRL n/a)

		signatures
Written by	STC SW Team	
Quality Assurance	Roberto Filippone	
Program Approval	Alessio Di Iorio	

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Change Log

Revision	Section	Changes
1.0	First Issue	

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Acronyms and Definitions

A	
AIT	Assembly & Integration Tests
AIV	Assembly & Integration Verification
ATP	Acceptance Test Procedure
ATR	Acceptance Test Report
B	
BOM	Bill Of Material
C	
CCB	Configuration Control Board
CM	Configuration Management
CoC	Certificato di conformità
COTS	Commercial Off-the-Shelf component
CP	Change Proposal
CP	Check Point
CSCI	Computer Software Configuration Item
CU	Control Unit
CDR	Critical Design Review
CFI	Customer Furnished Item
D	
DDR	Development Design Review
E	
ELT	Elettronica S.p.A. – Roma
EME	Electromagnetic Engineering
EMI	Electromagnetic Interference
EMC	Electromagnetic Compatibility
ESS	Environmental Stress Screening
F	
FAT	Factory Acceptance Test
FDR	Final Design Review
FoV	Field of View
J	
JSON	JavaScript Object Notation
H	
HMI	Human-Machine interface
IDD	Interface Design Description
I	
IRS	Interface Requirements Specification
L	
LoS	Line of Sight
LRU	Line-Replaceable Unit
M	
MoM	Minute of Meeting
MTBF	Mean Time Between Failures
N	
NC	Non Conformità
O	
OCD	Outline Control Drawing
OEM	Original Equipment Manufa
P	
PC	Personal Computer
PDR	Preliminary Design R
PDF	Portable Document Format
QP	Quality Plan
PM	Program Manager
PR	Problem Report
Q	
QM	Quality Manager
QMS	Quality Management System
QTP	Qualification Test Proc
R	
RR	Requirement Review
S	
SCP	System Change Proposal
SDD	Software Design Description
SPR	Software Problem Report
SRS	Software Requirements Specification
SSS	System Subsystem Specification
ST	Steering Turret
STCB	Steering Turret Control Boad
STC	Steering Turret CORAL
STD	Software Test Description
STR	Software Test Report
SUM	Software User Manual
SUT	System Under Test
SVD	Software Version Description
T	
TBC	To be Confirmed
TBD	To Be Defined
TN	Technical Note
V	
VTP	Verification Test Procedure
VTR	Verification Test Report
W	
WP	Work Package

Table 1 - Acronyms

SUT System Under Test. In the frame of the Steering Turret Coral, for SUT we intend the objects to be installed on the STC during operations. The SUT is a demo of the “missile seeker” that is an infrared trackers. The SUT operations is the scope the STC is built for. The SUTs are not part of the STC system.

FoV The field of view (FOV) is the angular extent of the observable world that is seen at any given moment. In the case of optical instruments or sensors, it is a solid angle through which a detector is sensitive to electromagnetic radiation. It is further relevant in photography.

LoS The line of sight, also known as visual axis or sightline (also sight line), is an imaginary line between a viewer/observer/spectator's eye(s) and a subject of interest, or their relative direction. The subject may be any definable object taken note of or to be taken note of by

the observer, at any distance more than least distance of distinct vision. In optics, refraction of a ray due to use of lenses can cause distortion.

Table 2 - Definitions

1 INTRODUCTION

1.1 SCOPE OF THE DOCUMENT

The SW Interface Control Document defines …...

1.2 IDENTIFICATION

This document is the SW Interface Control Document. It describes what does it describes? for the Steering Turret CORAL project – PHASE B). It is identified as ALMA-STC-123-ICD.

This document refers other documents that are part of the Requirement Review process, delivered in the same data package. All applicable and referenced documents are listed at § 2.

1.3 SYSTEM OVERVIEW

The STC is a complex system aimed to provide suitable support during countermeasure campaigns testing activities.

It is composed by:

- the **Control Unit**, allowing to process data and control all device equipment.
- the **Turret**, hosting the seekers and cameras
- the **Housing** cover and accessories, aiming to protect the device and workers during transport, storage and operations.

The Control Unit

The Control Unit will perform following main activities:

- The CU has to receive from the Coral Mainframe commands/messages needed for
  - FoVs Steering, LoSs Alignment and Lead Angles Application;
  - Cameras control;
  - Turret control;

  A joystick will be used to operate the Turret when disconnected from the Coral Mainframe.
- The CU has to send to the Coral Mainframe the messages needed for the monitoring of:
  - Steering angles, alignment angles and Lead angles;
  - Turret and Aiming Camera;
- The CU has to integrate a Control Panel in order to guarantee basic manual commands and provide information to the ST Operator;
- The CU has to host and to integrate the Tacking Board “CHARM-100NX” guaranteeing that it is able to:
  - receive in input the video frames coming from the Aiming Camera;
  - make available the tracking results to the CSCI ST Manager;
- The CU has to host and to integrate a board with adequate computing and storage resources for running the ST Manager CSCI.

The Turret

The Turret is designed to deploy eight seekers (SUTs) and two Cameras and to steer their FoVs towards a reference direction aiming flying platform targets. The Turret supports the following SUTs (simultaneously present) configurations:

- 6 SUT “heavy and compact”/”heavy and voluminous”+ 2 SUT “light and compact”;
- 8 SUT “light and compact”;
- 4 SUT “heavy and compact”/”heavy and voluminous” + 4 SUT “light and compact”.

The use of the STC involves:

- Seekers installation;
- Cameras and Seekers alignment;
- Operations to allow seekers calibration with Coral (video marker adjustments);

- Steering of SUTs and Cameras FoVs (Manually or Automatically) and Lead Angles application during target tracking.

The STC can be used independently or integrated with its external manager apparatus called Coral Mainframe.

The main functions of the STC are the following:

- FoV steering, the SUTs and Cameras FoVs are oriented in azimuth and elevation either in manual mode or, on the base of the "Target Tracking" output information, in automatic mode;
- Line of Sight (LoS) alignment, the SUTs and Cameras FoVs are angularly adjusted during the initial phase of their deployment on the ST, at the end of the alignment procedure the LoSs will be parallel each-other;
- Lead Angle Application, The Lead angle is the relative angle between the target direction and the SUT mechanical axis during its operation. Each SUTs has to be oriented in order to impose a Lead Angle;
- ST Control, the ST can be controlled either remotely (coming from either an external user or the Coral Mainframe) or through its Control Panel
- Target Tracking, while tracking the ST controls the azimuth and elevation of SUTs and Cameras FoVs in order to keep the flying target monitored;
- Self-levelling, the ST self-levels within specified range and accuracy on the horizontal direction;
- SUT protection, a suitable cover to protect the SUTs and Cameras from weather and weak intrusions is deployed;
- User Interface, an external user through HMI can visualize the state of the ST and send commands;
- SUTs cables connection, the cables directed to the SUTs are conveniently passed through the ST.

The Housing

The Housing has the function to provide full protection to the Turret from the weather conditions and to allow the safe transport.

The Housing also includes accessories and other devices to be used during deploy and operations. The System will be equipped with:

- Awning, to quickly apply a protection to the turret from the sudden rain;
- Lifting beam, to allow lifting of the Turret / Cover;
- Physical barrier, to protect the turret and the workers from possible accidents during operations;
- Mirror protection cover, tools, spare and any other equipment necessary to operate and keep a good level of efficiency.

1.4 DOCUMENT OVERVIEW

This document is composed by the following sections:

- **Chapter 1** "Identification", containing the general information on the document and a short description of the system it refers to.
- **Chapter 2** "Reference documents", listing the applicable and referenced documents;
- **Chapter 3** "**Chapter 3 title**".
- **Chapter 4** "Traceability Matrix", when applicable (not for this issue).
- **Appendices**, containing the appendices (if any).

2 REFERENCE DOCUMENTS

2.1 APPLICABLE

The documents listed in this section are considered to form part of this document. They are only applicable in so far as they are called up by individual sections of this document. For undated references, the latest edition of the document referred to shall be applicable.

Ref.	Document	Issue/rev - Date
[A.01]	Contract	4500069097/14114.05.2025
[A.02]	Statement Of Work for Development and Design of the Steering Turret	EDOC00335373 rev. 1 14.04.2025
[A.03]	Hardware Requirements Specification (HRS)	E0383388 rev 01 18.03.2025
[A.04]	Quality Requirements for Suppliers	EDOC00246137
[A.05]	Requisiti per i Fornitori ELT	Q0002501 DV rev 12

Table 3 - Applicable documents

2.2 REFERENCE

The documents listed in this section serve as reference material or possible sources of clarification and are for information only. For undated references, the latest edition of the publication referred to shall be consult.

Ref.	Document	Issue/rev - Date
[R.01]	QMS - Manuale di Qualità ALMA	ALMA-QMS-0001 v1.5 03/04/2025
[R.02]	MIL-STD-498	/ /

Table 4 – Reference documents

3 INTERFACE OVERVIEW

3.1 NETWORK SETTINGS

Le comunicazioni avvengono in UDP:

La ST assume il ruolo di Master

IP XXX.XXX.XXX.XXX porta 9966

La CU assume il ruolo di Slave

IP XXX.XXX.XXX.XXX porta 9914

3.2 USE OF JSON

JSON protocol has been selected for transmitting structured data between the Control Unit (CU) and the ST Control board (STCB).

JSON (JavaScript Object Notation), is a human-readable and compact solution to represent a complex data structure and facilitate data interchange between systems. It's a widespread data format with a diverse range of applications enabled by its simplicity and semblance to readable text. As such, it has been selected for communication between the Control Unit and the ST Control Board (STCB). The JSON is independent of the system's programming language, despite being derived from JavaScript and represents data that speaks common elements of many programming languages, effectively making it into a universal data representation understood by all systems.

……

The generic JSON message is composed as follow:

```
{ { Header },  "STATE": { State }, "TURRET": { Turret }, "MIRROR": { Mirrors }, "SUT": { SUTs }, "CAMERA": { Camera }, "LASER": { laserstate }, "SELFLEV": { SelfLevel }, "ERRORS": { ErrorList }, "HEALT": { HealtStatus } }
```

Forse non serve un header

All JSON stuctures described in the document have been validated through https://jsonlint.com/.

3.3 GENERAL

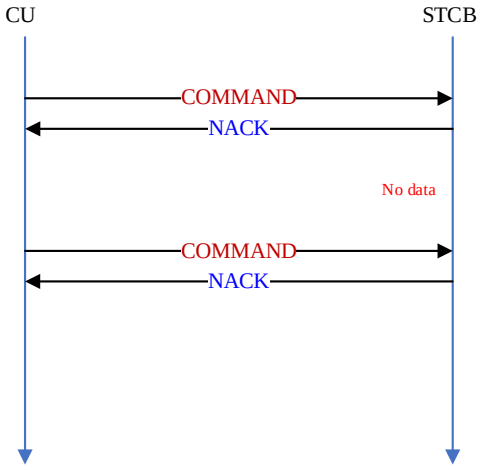


Figure 1 – Unknown user (no sysnc)

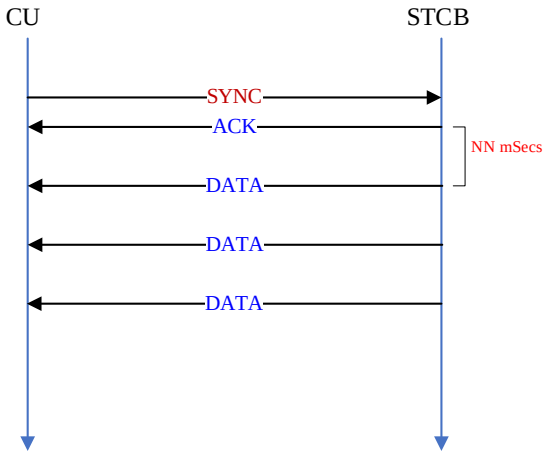


Figure 2 - CU - STCB sync

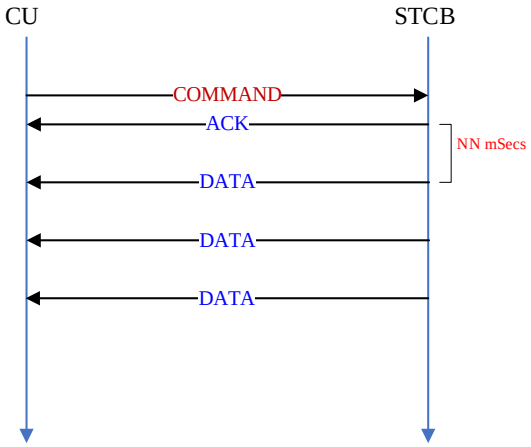


Figure 3 – COMMANDs from CU

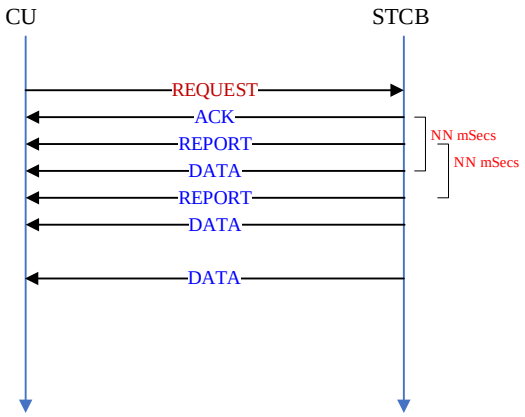


Figure 4 – Data request / Report

4 CONTROL UNIT – CORAL MAINFRAME

4.1 FROM CU TO CORAL MAINFRAME

4.1.1 DATA

4.1.1.1 Regular data

Definire un elenco minimale di dati da inviare con frequenza di XX mSecs. Il formato sarà TBD.

{ “DATA”: “binary data ??” }

4.2 FROM CORAL MAINFRAME TO CU

4.2.1 COMMANDS

4.2.2 RPTREQ – Report Request

5 CONTROL UNIT – ST CONTROL INTERFACE

5.1 FROM CU TO ST

5.1.1 COMMANDS

5.1.1.1 SYNC – Start communications

Viene inviato dalla CU come primo messaggio all'accensione. ST risponde con ACK. In seguito può essere inviato con cadenza regolare o secondo necessità.

ST esegue un controllo sugli elementi: l'interlocutore (**id**) deve essere conosciuto e la versione del protocollo utilizzato deve coincidere con quella interna in uso all'STBC.

```
{ "id": 11,"prot": "1.0a", "cmd": "SYNC" }
```

STBC accept commands only form application having one of the identifiers (**id**) specified in Table 5. After receiving SYNC the STCB start sending data regularly. The frequency (300 mSecs by default, can be changed via configuration file or can be dynamically changed by means of **DATAFREQ** (see 5.1.1.15.4).

SYNC will be accepted in any state.

id	*Description*
11	Control Unit
13	Logger application.
15	Diagnostic application
21	Simulator or other test applications

Table 5 – Application IDs

The version of the protocol matches with the last version (**1.0a**) of this document.

prot	*Description*
1.0a	The first issue of the protocol

SYNC is the command that initializes the communications with the connected application. According the application id and the protocol version, some information could be different.

5.1.1.2 STATE - Set State

Change the state of the STCB. Following states can be set by remote applications.

```
{ "STATE": "state" }
```

Following values will be accepted:

state	*Description*
INIT	La ST avvia le attività iniziali (self-test) e attende un successivo cambio di stato (OPER o MAINT)
OPER	La ST è operativa
MAINT	La ST è in manutenzione
FAIL	Se la CU è in FAIL e la torretta non può (o non deve) funzionare, anch'essa deve essere messa in FAIL. In generale il FAIL proviene dalla torretta stessa.

Table 6 – States of the STBC

5.1.1.3 SELFLEV - Start/Stop Self-level

Avvia/Ferma l'autolivellamento. La ST risponde con ACK, poi aggiunge lo stato del livellamento al successivo messaggio di stato (vedi § 5.2.2.2).

```
{ "SELFLEV": "value" }
```

value	*Description*
ON	Start the Self-Levelling
OFF	Stop the Self-Levelling
IBIT	Perform a Built In Test

Table 7 – Self levelling

During the self-levelling, the state and parameters of the self-level process is added to the regular telemetry.

5.1.1.4 STORAGE - Set Storage position

Set all motors in "storage position"

```
{ "cmd": "STORAGE" }
```

5.1.1.5 LASER - Set Laser ON/Off

Turn laser devices on/off

```
{ "LASER": "value" }
```

value	*Description*
ON	Start the Self-Levelling
OFF	Stop the Self-Levelling

Table 8 – Laser statesa

5.1.1.6 Move Turret - Azimuth

5.1.1.6.1 Relative

Muove la torretta di +/- gradi rispetto alla posizione attuale

5.1.1.6.2 To position

Muove la torretta ad una posizione fissa (gradi)

5.1.1.6.3 By speed

Muove la torretta alla velocità richiesta in direzione dx/sx.

5.1.1.7 Move Mirror(s) – Elevation

```
{ "MIRROR": "id", "angle": "angle" }
```

I valori di id sono elencati in Table 9; i valori di angle sono elencati in Table 11.

1	Motore specchio 1	
2	Motore specchio 2	
8	Motore specchio 8	
All	Per tutti i motori specchio	

Table 9 – Mirror motor selection id

5.1.1.7.1 Relative

Muove il motore specchio di +/- gradi rispetto alla posizione attuale.

angolo	step	Risoluzione minima +/- 0.01°

Table 10 – Mirror angle

5.1.1.7.2 To position

Muove il motore specchio ad una posizione fissa (gradi).

angolo	-10.00 … +16.00 step 0.01	

Table 11 – Mirror angle

5.1.1.7.3 By speed

Muove il motore specchio alla velocità richiesta in direzione dx/sx.

speed	Angolo/sec	Risoluzione minima +/- 0.01° / sec

Table 12 – Mirror speed

5.1.1.8 Move SUT(s) - Azimuth

5.1.1.8.1 Relative

Muove il motore SUT di +/- gradi rispetto alla posizione attuale.

5.1.1.8.2 To position

Muove il motore SUT ad una posizione fissa (gradi).

5.1.1.8.3 By speed

Muove il motore SUT alla velocità richiesta in direzione dx/sx.

5.1.1.9 Move Camera - Elevation

5.1.1.9.1 Relative

Muove il motore Camera di +/- gradi rispetto alla posizione attuale.

5.1.1.9.2 To position

Muove il motore Camera ad una posizione fissa (gradi).

5.1.1.9.3 By speed

Muove il motore Camera alla velocità richiesta in direzione dx/sx.

5.1.1.10 RESET – Reset all devices to default values

Reset all devices to default values

```
{ "cmd": "RESET" }
```

5.1.1.11 DELLOG - Delete log file(s)

Deletes all log and temporary files.

```
{ "cmd": "DELLOG" }
```

5.1.1.12 FACTORY – Set factory settings

Set to factory settings, including work time counter.

```
{ "cmd": "FACTORY" }
```

5.1.1.13 SELFTEST – Perform the ST Self Test (IBIT)

Request a complete self test

```
{ "cmd": "SELFTEST" }
```

5.1.1.14 SHUTDWN - Shut-down

Immediate shutdown of the system

```
{ "cmd": "SHUTDWN" }
```

5.1.1.15 Change config

Per adesso queste funzioni non le implementiamo se non sono necessarie. Però il file di configurazione del SW sulla STCB li prevede, e possono esse impostate manualmente editando il file default.cfg

5.1.1.15.1 BARRIER - Enable/disable physical barrier

5.1.1.15.2 LAMP - Enable/disable lamp

5.1.1.15.3 BEEPER - Enable/disable beeper

5.1.1.15.4 DATAFREQ - Change Data frequency

This command allows the connected application to change the frequency of the telemetry data. This can be useful to increase frequency during critical phases of processing and make it slow during other phases (e.g. stand-by).

The frequency change is not permanent; At application restart, the default frequency value will be applied. To permanently change the telemetry frequency, change the STCControl config file.

```
{ "DATAFREQ": "frequency" }
```

frequency	*Description*
0	Disable telemetry
10…… 3000	Telemetry frequency, in milliseconds. Min value 10 mSecs, max 3000 mSecs (= 3 secs)

Table 13 – Telemetry frequency

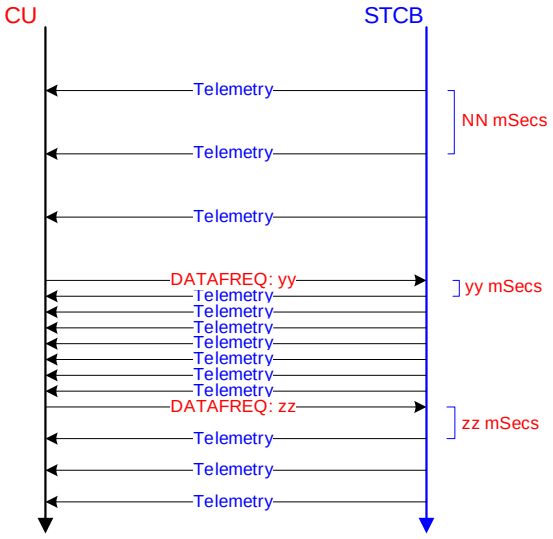


Figure 5 – Change telemetry frequency

5.1.1.15.5 FORCEPOS – Force a specific position/heading

This command is used for test scopes. It ask the turret to ignore the GPS position and assumes the position and heading specified in the command. The internal GPS will be ignored until a RESTPOS will be received.

```
{ "DATAFREQ": [latitude, longitude, heading] }
```

es.:

```
{ " FORCEPOS": [41.890210, 12.492231, 45.16] }
```

5.1.1.15.6 RESTPOS – Restore the internal GPS

This command is used for test scopes. It asks the turret to restore the use of the internal GPS receiver.

```
{ "cmd": "RESTPOS " }
```

5.1.2 RPTREQ – Report Request

Report request. Following reports are available for request:

- Log files (date)
- Current config file
- Current errors
- Full state
- SW version of STCB
- Total time of working

See

LOG	TODAY	Return the requested log file .
	yyyymmaa	Log of day dd/mm/yyyy

for possible values of repname

```
{ "rpt": "repname" }
```

repname	*param*	
CONFIG		Return the configuration file used by the ST Control application
ERRORS		Return the list of active error codes
FULL		Complete ST status (see § 5.2.3)
SWVER		Firmware and OS version
WRKTIME		Return the time elapsed since the first run
LOG		Return the log file of the current day

Table 14 – Report request

The next version of the document will implement:

```
{ "rpt": { "LOG": "param" } }
```

repname	*param*	
LOG	TODAY	Return the requested log file .
	yyyymmaa	Log of day dd/mm/yyyy

La ST risponde con il report richiesto (vedi § 5.2.2, § 5.2.3, § 5.2.4)

5.2 FROM ST TO CU

5.2.1 Acknowledge

STCB replies ACK to all commands.

{ "id": 11,"prot": "1.0a", "ACK": "code" }

Vedi Table 15 per i valori di code:

code	*Description*	*Note*
00	No error	
01	Client unknown	
02	Protocol mismatch	
03	Syntax error	
04	Value out of range	
05	Not allowed in this state	

06	Insufficient resources	N/A in this version of the document

Table 15 - ACK codes

5.2.2 DATA

5.2.2.1 Regular data

Definire un elenco minimale di dati da inviare con frequenza di 10 mSecs. Il formato sarà binario.

{ “DATA”: “binary data” }

5.2.2.2 SelfLevel

Following data will be sent to the CU only during the self-levelling phase or on request (IBIT).

```
{ "SELFLEVEL": "on|off", "XAxis": "-10.00|10.00", "YAxis": "-10.00|10.00", "MOTORS":
[ "on|off", "on|off", "on|off", "on|off" ] }
```

- Value of SELFLEVEL state

code	*Description*	*Note*
on	Self-leveling is running	
off	Self-leveling is not running	

- Values of Xaxis and YAxis range from -10.00 to 10.00 (resolution 0.1)

- Value of four MOTORS, identified in clock-wise order as M_1 (front-left), M_2 (front right), M_3 (back right), M_4 (back left), respect to the front of the Turret. Each Motor will have one of following states:

code	*Description*	*Note*
on	Motor M_n is running	with n = 1,…4
off	Motor M_n is not running	with n = 1,…4

5.2.2.3 Storage

La torretta è in posizione di storage (può essere spenta)

5.2.2.4 BarrierBreak

Se la barriera di protezione è stata infranta, informa la CU. Il sistema va in protezione (Quikly shut-down).

Forse riusciamo ad avere un allarme intermedio ad una prima distanza.

5.2.2.5 TurretMotor

5.2.2.6 MirrorMotor

5.2.2.7 SUTBreak

5.2.2.8 CameraBreak

5.2.2.9 Laser

5.2.3 Full state

Restituisce:

- Stato del sistema
- Versione del SW
- Versione del Sistema Operativo
- Tempo di lavoro
- Temperatura/umidità/Pressione
- Posizione (lat/lon) e orientamento della torretta;
- Posizione attuale di tutti i motori (o STORAGE)
- Direzione e velocità attuale di tutti i motori
- Stato di abilitazione dei gruppi SUT/Specchio
- Stato on/off di tutti i laser;
- Stato completo del sistema di livellamento;
- Elenco di tutti gli errori attuali.

5.2.4 REPORT

5.2.4.1 CONFIG Log file

5.2.4.2 ERRORS Config file

5.2.4.3 FULL Error list

5.2.4.4 SWVER

```
{ "rpt": " SWVER", "nrec": 1, "trec": 1, "rec": "record content" }
```

5.2.4.5 WRKTIME

5.2.4.6 LOG

5.2.5 HEALTH

5.2.5.1 Error list

000		
001		

Table 16 - Error codes

- The HEALT is a list of error codes, currently active in the Self-Level management (errors may come from the Self-level device or from the STCB due to interface problems). Following is the full list of possible values:

code	*Description*	*Note*
TBD	Generic error	

TBD	SL incomplete	
TBD	Comm error	
TBD	Excessive slope	
TBD	Motor fault	

6 SELF-LEVEL MANAGEMENT

The Self-level system is composed by following devices:

- four motors, identified in clock-wise order as M1 (front-left), M2 (front right), M3 (back right), M4 (back left), respect to the front of the Turret;
- a two-axis inclinometer;
- a controller, connected to the ST-MCU via RS-232.

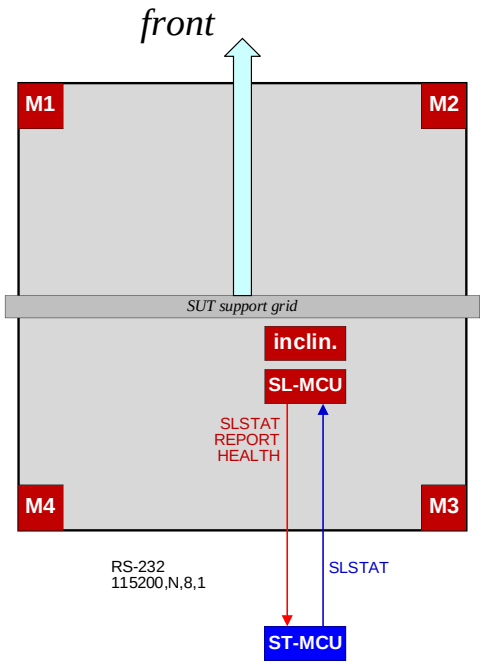


Figure 6 – The Self-Level sub-system

The communication between the Self-Level system and the Steering Turret system will be performed via RS-232 serial communication. The RS setting is:

baudrate: 115200

parity: None

data: 8

stop: 1

No CTS.

On the Steering Turret side, the ST-MCU application will work as Master (it is the communication initiator);

On the Self-Level side, the SL-MCU application will work as Slave and reply to ST-MCU request.

Following is the general structure of the message:

ID	data	LRC

ID is the message descriptor. It is at the beginning of the frame. Following table show the possible values of ID:

Acronym	*Hexa value*	*Bin value*	*description*	*Data bytes*
SLSTAT	0xF0	11110000	Master: order to change the status of the SL	1

			Slave: return current status of the SL	
REPORT	0xF3	11110011	Slave: return the angles and motor modes	6
HEALTH	0xF6	11110110	Slave: return the health status of the devices	TBD

The LRC is a one-byte Longitudinal Redundancy Check suitable for short serial communication messages. It is a computationally lightweight XOR-based.

LRC = data[0] ^ data[1] ^ ... ^ data[n-1]

6.1 SLSTAT – SELF LEVELLING STATUS.

The ST-MCU request a SL change status. The SL-MCU replies with the new state, or HEALTH in case of error

cmd = 0xF0

data (1 byte) = Start/Stop SL/status request, as follow:

Hexa value	*Description*	*Note*
0x01	Start	Master: start self-level Slave: the self-level is ON
0x02	Stop	Master: stop self-level Slave: the self-level is OFF
0x04	Report Request	Master: full status request Slave: N/A

Table 17 – table

6.2 REPORT – SELF LEVELLING DATA.

This command is provided by SL-MCU only, every 300 mSecs (TBC), for the whole period of self levelling. When self-levelling is complete, the SL-MCU sends a SLSTAT.

cmd = 0xF3

data (6 bytes) = full status data as follow:

Byte	*name*	*Description*	*Note*
1	SL status	See SLSTAT data	The status of the SL (ON/OFF)
2	Motor status	See Motor Status	1 byte (bit based) including the status of each motor (ON/OFF)
3	X angle	Inclination of X	Integer part (-10 / +10)
4	X angle		Decimal part (1…99)
5	Y angle	Inclination of Y	Integer part (-10 / +10)
6	Y angle		Decimal part (1…99)

Table 18 – table

Motor Status

1 byte including the motor status in the bit 0…3, as follow (bit 0 is the LSB, bit 7 is the MSB):

Bit	*name*	*Description*	*Note*
0	M1	1 = M1 is ON - 0 = M1 is OFF	
1	M2	1 = M2 is ON - 0 = M2 is OFF	
2	M3	1 = M3 is ON - 0 = M3 is OFF	
3	M4	1 = M4 is ON - 0 = M4 is OFF	
4	n/a	Must be zero	
5	n/a	Must be zero	
6	n/a	Must be zero	
7	n/a	Must be zero	

Table 19 – table

Inclination

Inclination values range is from -10.00° to +10.00°.

Integer part of the inclination angle has the MSB high for negative values: eg:

+10° => 0x0A

-10° => 0x8A

decimal part of the inclination values ranges from 00 to 99

6.3 HEALT – SELF LEVELLING SYSTEM HEALTH STATUS

The health status will be sent on report request and any time an error/failure occurs during the self level.:

cmd = 0xF0

data (6 bytes) = health status data as follow:

Byte	*name*	*Description*	*Note*
1	FW version	The firmware version	
2	Reserved	Reserved for future scopes	
3	Health byte 1	TBD	
4	Health byte 2	TBD	
5	Health byte 3	TBD	
6	Health byte 4	TBD	

FW version: one byte that specifies the Self-Level FW version;

Reserved: reserved for future scopes;

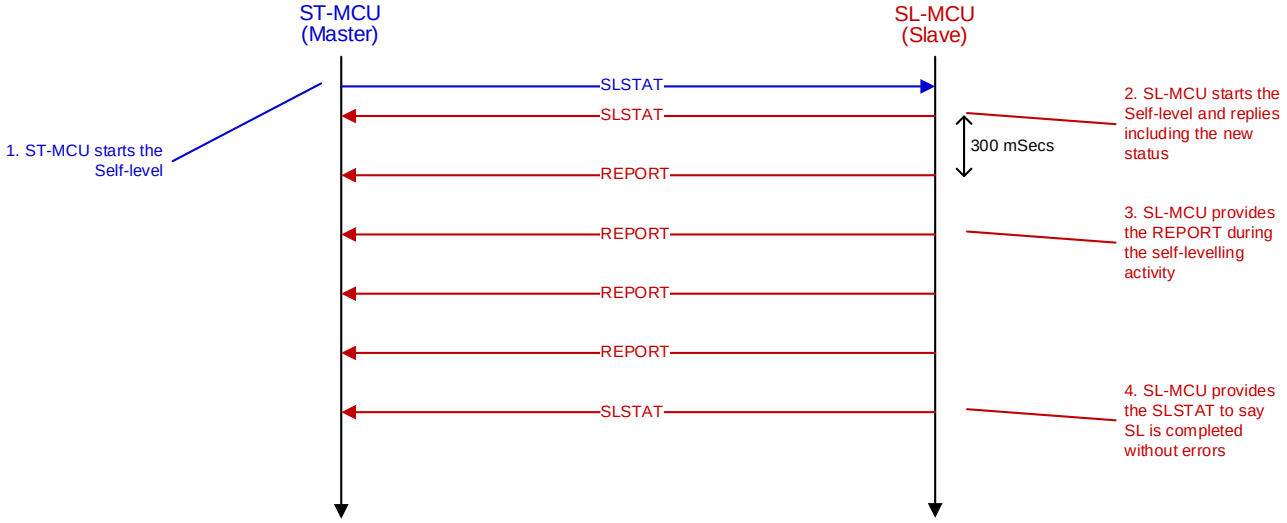
Health byte: 4 bytes including the full health status (processing errors/failures). The use of each bit is still TBD.

6.4 SEQUENCES

All sequences start and finish with a SLSTAT.

6.4.1 Nominal processing

ST-MCU start the Self-Level. SL-MCU performs the self-level without errors.



SLSTAT	start	
0xF0	0x01	*LRC*

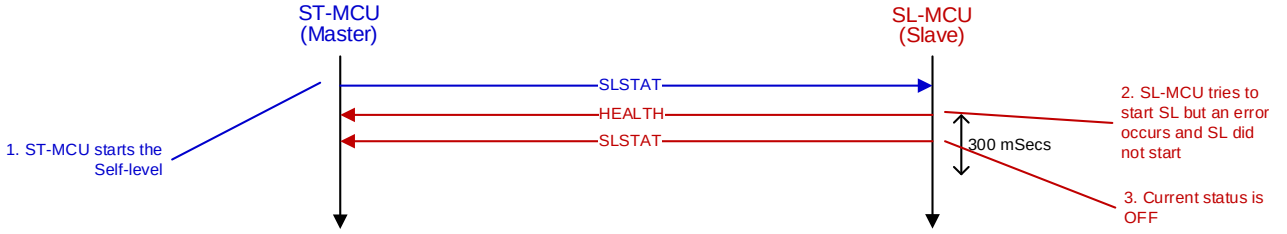
SLSTAT	started	
0xF0	0x01	*LRC*

REPORT	SL ON	M3 ON	Inclination X (+3.10°)		Inclination Y (-2.7°)		
0xF3	0x01	0x04	0x03	0x0A	0x82	0x46	*LRC*

SLSTAT	end	
0xF0	0x02	*LRC*

6.4.2 Self-Level impossible

ST-MCU start the Self-Level. SL-MCU cannot perform the SL due to an error or the platform inclination is out of range.



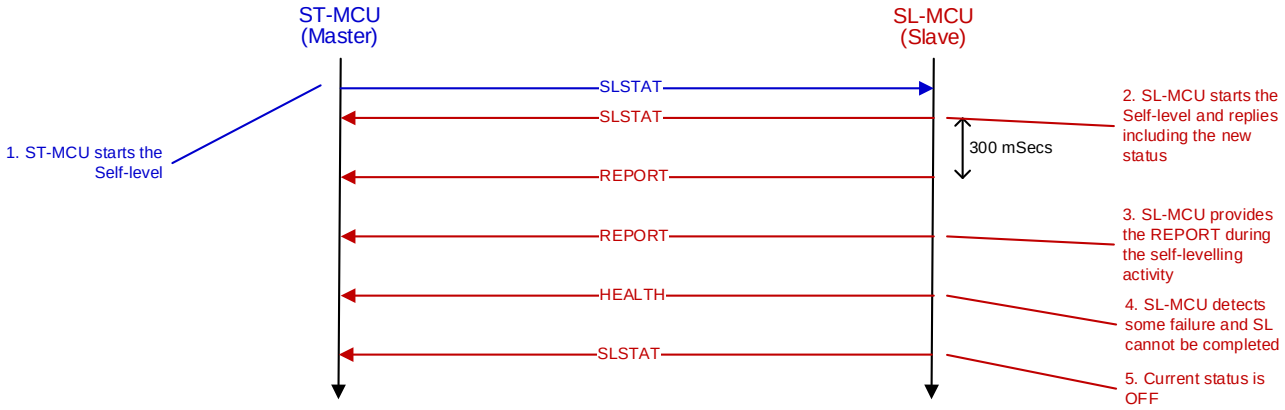
SLSTAT	start	
0xF0	0x01	*LRC*

HEALTH	H Byte 1	H Byte 2	H Byte 3	H Byte 4	
0xF6					*LRC*

SLSTAT	end	
0xF0	0x02	LRC

6.4.3 Failure during Self-level

ST-MCU start the Self-Level. SL-MCU perform the SL but after a while it detects some failure and SL cannot be completed



SLSTAT	start	
0xF0	0x01	LRC

SLSTAT	started	
0xF0	0x01	*LRC*

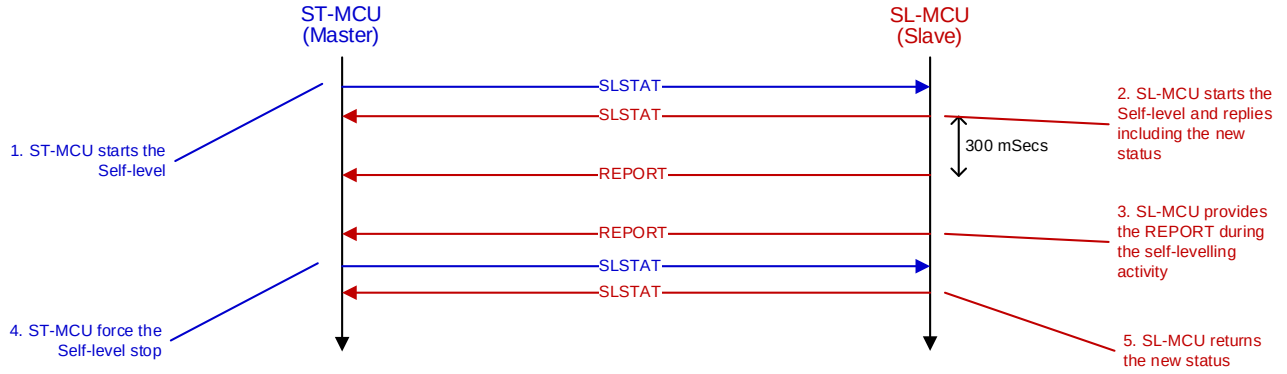
REPORT	SL ON	M3 ON	Inclination X (+3.10°)		Inclination Y (-2.7°)		
0xF3	0x01	0x04	0x03	0x0A	0x82	0x46	*LRC*

HEALTH	H Byte 1	H Byte 2	H Byte 3	H Byte 4	
0xF6					*LRC*

SLSTAT	end	
0xF0	0x02	*LRC*

6.4.4 Stop request during Self-level

1. ST-MCU start the Self-Level. SL-MCU perform the SL but after a while the ST-MCU request for a stop. The SL is not completed but SL-MCU return the current new status.



SLSTAT	start	
0xF0	0x01	*LRC*

SLSTAT	started	
0xF0	0x01	*LRC*

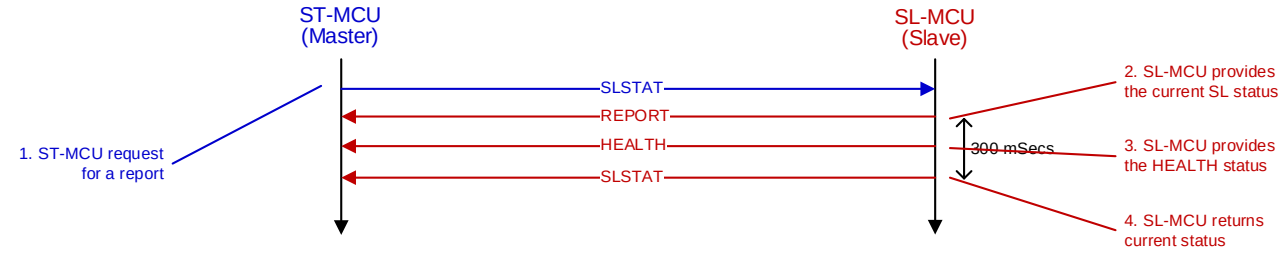
REPORT	SL ON	M3 ON	Inclination X (+3.10°)		Inclination Y (-2.7°)		
0xF3	0x01	0x04	0x03	0x0A	0x82	0x46	*LRC*

SLSTAT	stop	
0xF0	0x02	*LRC*

SLSTAT	end	
0xF0	0x02	*LRC*

6.4.5 Report request

ST-MCU start its own self-test/IBIT. SL-MCU returns the current status (REPORT) and the health status (HELTH).



SLSTAT	report	
0xF0	0x04	*LRC*

REPORT	SL OFF	All motors OFF	Inclination X (+0.01°)		Inclination Y (-0.02°)		
0xF3	0x00	0x00	0x00	0x01	0x80	0x02	*LRC*

HEALTH	H Byte 1	H Byte 2	H Byte 3	H Byte 4	
0xF6					*LRC*

SLSTAT	end	
0xF0	0x02	*LRC*

7 DISCRETE SIGNALS

7.1 SAFETY/ARM

8 DEVICE CONTROL

8.1 CANBUS

8.1.1 Turret motor

8.1.2 Mirror motor

8.1.3 SUT motor

8.1.4 Camera motor

8.1.5 Environmental sensors

8.1.6 GPS heading

9 TRACEABILITY MATRIX

N/A to the present issue of the document.

A.1 LIST OF COMMANDS

A.1.1 Appendix

cmd	*state*	*Parameters*	*Description*
SYNC	any		Start comminucations
STATE	any	INIT	Set the new ST state
		OPER	
		MAINT	
		FAIL	
SELFLEV	INIT	start	Start / stop selef level
	MAINT	stop	
STORAGE	OPER		Set the Turret in storage position
	MAINT		
LASER	OPER	on	Switch the laser on/off
	MAINT	off	
RESET	any		Reset all devices to default values

END OF DOCUMENT

ALMA Sistemi srl
P.IVA/VAT nr. 08390011008 - R.E.A. RM 1091337
Tel: +39 (0)6 95550355
E-mail: info@alma-sistemi.com
PEC: alma_sistemi@pec.it
Web: www.alma-sistemi.com

Legal office
4, Via Dei Nasturzi
00012 - Guidonia Montecelio (Rome) - Italy

Headquarter
Via Giacomo Peroni, 400 Building 3/B, 1st floor
000131 - Rome - Italy

Secondary office
Strada Statale 17 bis, Km 41.970 snc
67100 - Bazzano (L'Aquila) – Italy

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